Introduction to NumPy

Agenda

- Fundamentals of NumPy
  - Advantages of NumPy
  - NumPy: Installation and Import
- NumPy: Array Object
  - Creating NumPy Arrays

1. Fundamentals of NumPy

NumPy (Numerical Python) is a free and open-source library that is mostly used for mathematical operations in scientific and engineering applications.

- It is a Python library used for working with arrays.
- It consists of a multidimensional array of objects and a collection of functions for manipulating them.
- It conducts mathematical and logical operations on arrays.

Note: The array object in NumPy is called ndarray.

1.1 Advantages of NumPy

- It provides an array object that is faster than traditional Python lists.
- It provides supporting functions.
- Arrays are frequently used in data science.
- NumPy arrays are stored in one continuous place in memory unlike lists.

1.2 NumPy: Installation and Import

- `C:\Users\Your Name>pip install numpy` command is used to install NumPy.
- NumPy is imported under the name np like `import numpy as np`

2. NumPy: Array Object

A NumPy ndarray object can be created by using the array() function.

```
# Example:
import numpy as np
arr = np.array ([10,20,30,40,50])
print (arr)
print (type(arr))
```

```
[10 20 30 40 50]
<class 'numpy.ndarray'>
```

2.1 Creating NumPy Arrays

- Create multiple dimensional arrays, such as 0D, 1D, 2D, and 3D

```
# Create a 0D Array
arr0 = np.array(24)
print ('0D array is', arr0)
# Create a 1D Array
arr1 = np.array([1,2,3,4])
print ('1D array is', arr1)
# Create a 2D Array
arr2 = np.array([[1,1,1],[1,2,1]])
print ('2D array is', arr2)
# Create a 3D Array
arr3 = np.array([[[1,1,1],[2,2,2]],[[3,3,3],[4,4,4]]])
print ('3D array is', arr3)
```

```
0D array is 24
1D array is [1 2 3 4]
2D array is [[1 1 1]
 [1 2 1]]
3D array is [[[1 1 1]
  [2 2 2]]

 [[3 3 3]
  [4 4 4]]]
```

Observations:

- 1D arrays are unidimensional and have 0D arrays as its elements.
- 2D arrays have 1D arrays as their elements.
- 3D arrays have 2D arrays as their elements.